Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Embedded Strain and Temperature Sensing in Additively Manufactured Smart Metallic Structures			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), **Advanced Mobility (√), Next-gen Nuclear Power (√),** Advanced Bio ( ), **Aerospace/Marine (√),** Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), **Advanced Robotics & Manufacturing (√),** Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 1** (Large Category)	
	Mechanical and Materials		Mechanical and Materials	
	Medium Category	**Small Category**	**Medium Category**	**Small Category**
	Metal Materials	Structural Materials Functional Materials Composite Materials Materials Processing Technologies Materials Analysis	Robotics/Automation Machinery	Measurement/Sensor Technologies for Automation
			Shipbuilding/Marine Systems	Ship Materials/Structural Technologies
			Aerospace Systems	Other Aerospace System Technologies
	Technology Field 2 (Large Category)		**Technology Field 3** (Large Category)	
	Energy and Resources		Knowledge Services	
	Medium Category	**Small Category**	**Medium Category**	**Small Category**
	Nuclear Power	Nuclear-Based Advanced Technology	R&D/ Engineering Services	Testing/Inspection/Analysis Technique
Concept of Proposed Technology	**Project Overview:** In this project, as illustrated in Figure 1a, we propose the **additive manufacturing** of **Smart Metallic Structures (SMS)**[1] featuring **embedded sensing capabilities** for **structural health monitoring (SHM)**[2]. These structures are intended for deployment across a range of application domains, including **aerospace**[3], **marine**[4], **civil infrastructure**[5], and **nuclear power**[6] (see Figure 1b).			

- Building on a **US patent-pending process** (Fig. 1a) developed through a recently concluded **DARPA**-funded project[7], the RPI team has established a novel manufacturing capability for embedding multiple-sensing modalities (i.e., strain, temperature, piezoelectric transducers) directly within additively manufactured metal structures.
- The process integrates **wire-arc additive manufacturing (WAAM)**, targeted **subtractive machining, ceramic shielding,** and **selective metallization** to enable seamless deposition over the instrumented regions, while maintaining electrical continuity and structural integrity of the commercial off-the-shelf (COTs) sensors.
- Using this approach, we have embedded sensors in **Al-5356 WAAM-build structures**, achieving high-fidelity temperature and strain field measurements with minimal mechanical debit and sensor stack survivability under demanding deposition conditions.

Therefore, in this joint TIPA proposal, we bring **a platform capability—not just a single device** and have already demonstrated sensor survivability, structural performance, and measurement accuracy under realistic processing conditions.

The proposed effort will move RPI's IP demonstration (Figure 1a) beyond its initial aluminum WAAM implementation to **metals and geometries that are of interest to Korean SMEs**. With the Korean SME partners defining the target application and operating envelope, the project aims to advance the technology from its current **TRL 3** toward **TRL 5 system-relevant prototypes**.

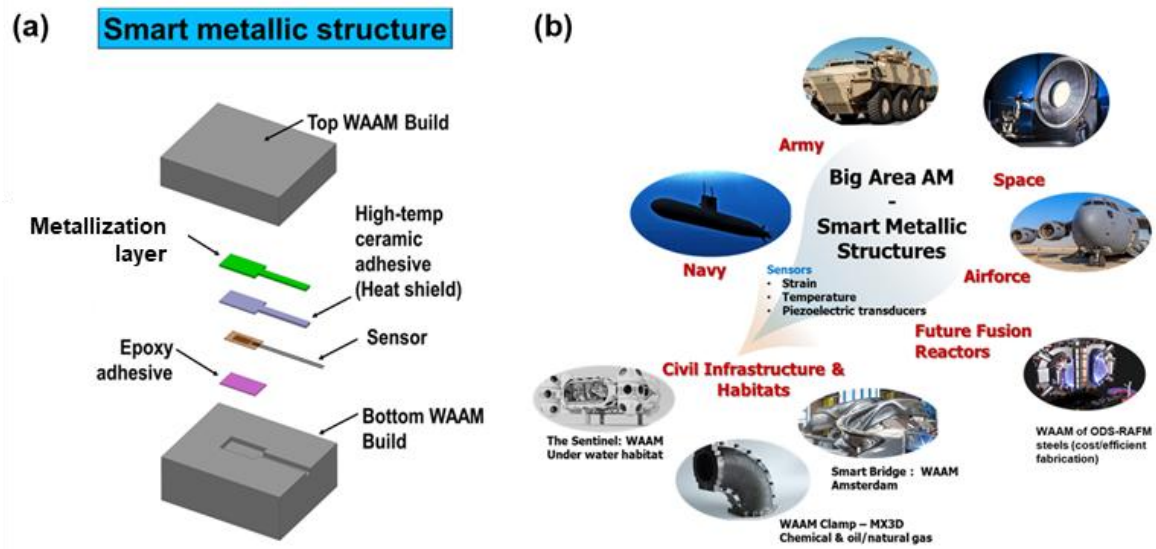

Fig. 1 (a) RPI's patent-pending convergent manufacturing approach for sensor-embedding into metallic structures; (b) Potential application domains of RPI's convergent manufacturing platform.

Reduction to Practice – Additively Manufactured Smart Beam Prototypes: RPI's demonstrations to-date have been performed on a WAAM bending beam (180 mm × 20 mm × 13 mm) incorporating a **two-layer embedded sensor network** (Figure 2a). Each layer contained **four RTDs and three strain gauges**, arranged to capture through-thickness temperature gradients and bending-dominated strain fields with high spatial resolution. Strain gauges were embedded asymmetrically with respect to the neutral axis to evaluate gradient sensitivity and measurement fidelity. Sensors were spot-welded to a flexible printed circuit board (FPCB), bonded to the pre-treated WAAM surface, and subsequently encapsulated using ceramic shielding and conductive metallization. Side-tab recesses machined after WAAM resumption exposed the FPCB leads for electrical interfacing (Figure 2b), enabling reliable connection to external data-acquisition hardware while preserving structural integrity. This packaging and routing strategy ensured robust electrical continuity, mechanical anchoring, and survivability through all embedding steps. Figure 2b showcases the real-time strain response measured by the embedded strain gauges under the applied load.

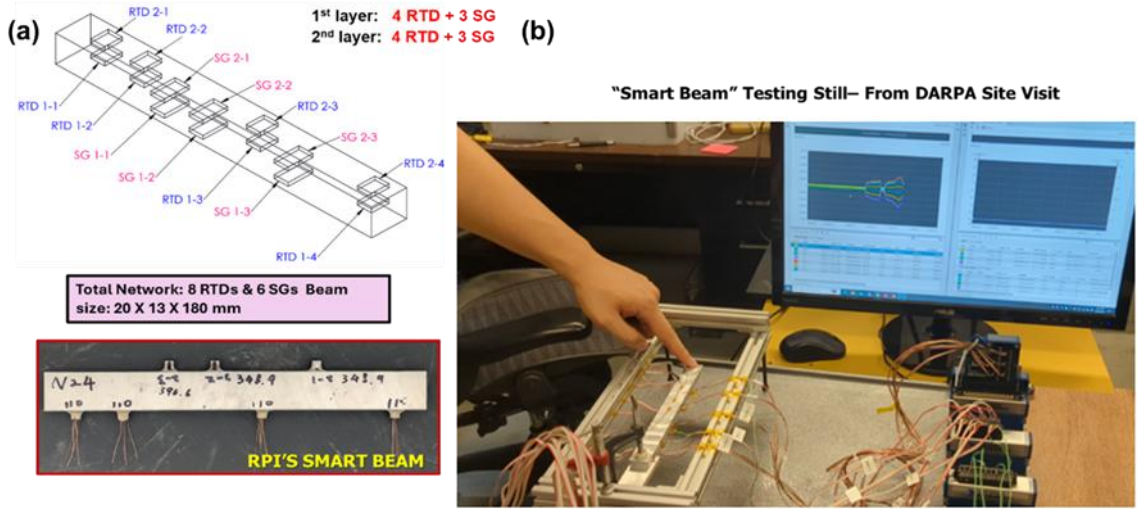

Fig.2 (a) Smart Beam design featuring two layers of sensor networks (each layer includes 4 RTDs and 3 strain gauges (SGs), for a total of 14 embedded sensors) (left panel) and (b) Still from live testing demonstration showing embedded sensing in cantilever configuration (right panel).

Microstructural evaluations further confirmed the robustness of the sensor embedding process (Figure 3). Cross-sections taken in transverse (A–A) and longitudinal orientations revealed **dense WAAM deposition with no delamination, continuous metallurgical bonding** at the WAAM–metallization layer interface, and tight ceramic conformity around the embedded sensors. Optical imaging has confirmed uniform metallization, strong WAAM–metallization layer adhesion, and intact FPCB traces and adhesive layers, demonstrating that thermal and mechanical loads during WAAM deposition did not compromise sensor integrity.

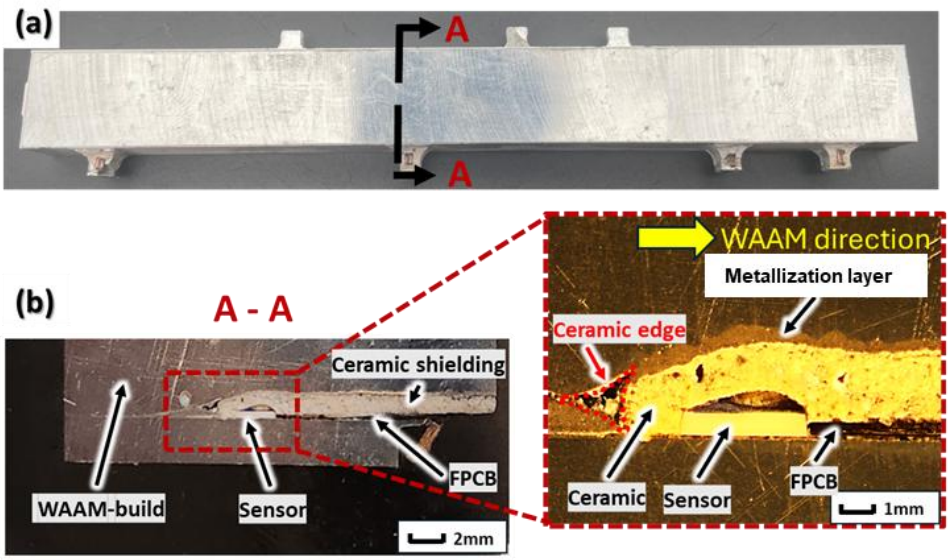

Fig.3 Microstructural analysis of the Smart Beam: (a) Beam image with regions of interest; (b) Optical microscope image of Section A–A showing the WAAM (top layer)–ceramic surface metallization and bottom layer WAAM interfaces.

Early-Stage Testing Results from Select RPI Prototypes:

- Thermal and mechanical evaluations were performed on Smart Beams incorporating embedded temperature sensors (RTDs) and strain gauges. The thermal tests included steady-state and transient heating. These results <u>demonstrate spatial–temporal sensing capability across the embedded RTD network</u>.

- Static three point bending tests were also conducted to evaluate strain sensing. Embedded gauges recorded flexural deformation across the span, and reconstructed strain histories matched surface gauge measurements with relative errors below 0.5%. The embedded

	sensors maintained a linear response throughout the elastic loading range, providing reliable through thickness strain information. These preliminary results have validated the feasibility of embedding multi-layer sensor networks within additively manufactured metallic structures
Rationale for Joint R&D Project	A joint R&D project is justified for the following reasons: • **Industry-defined application and acceptance criteria.** While RPI's coupon-level demonstrations establish process feasibility, they do not define the method's operating envelope. The industry partner will select the target component, define the operating environment, and specify sensing and acceptance criteria - ensuring development is application-driven rather than speculative. • **Qualification-grade data.** Industrial adoption requires reproducible thermal, static, dynamic, fatigue, and microstructural datasets with clear pass/fail criteria. Joint R&D is essential to generate test matrices, statistically meaningful datasets, and benchmark comparisons. • **Manufacturability and scale-up validation.** Transitioning from lab-scale coupons to subcomponents requires validation of cycle time, yield, process windows, and key cost drivers. • **Co-development (not licensing).** This is explicitly a co-development effort: the industry partner will help define performance requirements, test protocols, and prototype specifications. This model aligns with Korean SME R&D practices and accelerates targeted TRL advancement. • **TRL advancement with application relevance.** The project targets **TRL 3 → TRL 4 → early TRL 5** by validating the patent-pending embedding approach for application-relevant materials, geometries, and operating environment.
Key Words	1. Additive Manufacturing (AM) 2. Wire-arc additive manufacturing (WAAM) 3. Powder direct-energy deposition (DED) 4. Embedded Multimodal Sensing 5. Structural Health Monitoring (SHM)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3)	**End of the Project**	Level (5)
R&D Objectives	The proposed R&D will mature RPI's patent-pending, convergent manufacturing approach into industry-relevant prototypes by pursuing the following objectives: • **Co-define application and acceptance criteria:** Korean partner-led selection of target component, operating envelope, sensing specification, and quantitative acceptance thresholds to ensure application-driven technology development. • **Adapt processes and materials for platform portability:** Refine the WAAM/DED + subtractive machining + ceramic-shielding + conductive metallization sequence by down-selecting material systems and establishing robust process windows for the process. • **Integrate and qualify multilayer, multimodal sensor networks:** Develop encapsulation, routing, and readout architectures for piezo sensors, RTDs and strain gauges; validate electrical continuity, EMI/thermal protection, and signal integrity under representative conditions. • **Fabricate application-relevant prototypes and perform qualification testing:** Produce prototypes and execute thermal, static, dynamic, fatigue, and microstructural tests to quantify sensing fidelity, mechanical debit, and durability against industry benchmarks. • **Advance TRL and define the transition pathway**: Identify remaining qualification gaps and			

produce a co-defined roadmap to move the capability from TRL-3 toward TRL-5 with partner demonstrations.

R&D Target Specifications

➔ The project aims to achieve high-precision, three-dimensional sensing of temperature and strain fields in additive manufactured metal structures - under demanding thermal and mechanical conditions, while simultaneously preserving structural and sensor integrity.

➔ Measurement performance will be validated using controlled thermal and mechanical gradients applied to the specimens, with comparative benchmarks established against sensor-free baseline structure. The performance targets envisioned include:

Metric	Units	Target
Temperature accuracy	°C	± 20°C
Strain accuracy	ε(m/m)	± $5x10^{-4}$
Static property debit	GPa/MPa/%	< 10%
Sensor durability	hours	100
Dynamic property debit	Cycles to failure	< 20%

These specifications represent our internal performance objectives and will be refined in partnership with Korean small and medium-sized enterprises (SMEs) to address application-specific needs.

R&D Plan

Task 1 – Application Geometry and Materials definition (Industry-led)

Task 1 will establish the application-specific foundation that guides all subsequent technical work. The industry partner will identify the target component geometry and operational envelope, and will define the sensor modality, sensor network architecture, and sensing requirements needed to meet performance, durability, and diagnostic objectives for the proposed smart metallic structure. This co-definition effort will also encompass selection of the relevant metallic system(s) (Note: RPI's patent-pending sensor embedding technology has already been demonstrated in structural aluminum).

From the RPI side, both wire-arc additive manufacturing (*WAAM; Fronius-CMT*) and laser powder directed energy deposition (*DED; FormAlloy*) platforms are available and will be leveraged to support additive manufacturing, prototyping (Figure 4), and materials evaluation activities required in the project. Allowable component geometries - and the final selection of sensor types and their placement within the structure- will be finalized in coordination with, and ultimately defined by, the industry partner.

Fig 4. RPI's wire-arc additive manufacturing (WAAM), and rapid alloy development via Form Alloy laser-DED capabilities.

Task 2 - Process and materials adaptation (university-led)

This task will establish the convergent manufacturing pathway required to embed functional strain and temperature sensors within metallic and advanced alloys (e.g., high-entropy alloy) structures. Building on the RPI team's prior experience with WAAM-based fabrication and sensor-integration methods, the proposed work will address the key processing steps that govern thermal management, metallurgical compatibility, electrical continuity, and structural integrity during sensor embedding. **The goal is to develop a robust, transferable embedding sequence that can be applied across structural metal alloys**. Note: Advanced alloys (i.e., HEAs, functionally graded materials) are also feasible by using the multi-axis powder-DED machine (*FormAlloy laser-DED*), along with its dedicated powder feeder.

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Sub-Task 2.A – AM Host Structure Fabrication and Surface Preparation. This sub-task will establish the baseline AM process for fabricating the host metallic structure (Fig. 5-Step 1). The team will optimize deposition parameters, bead geometry, and interlayer thermal control to ensure consistent layer formation and strong metallurgical bonding. Surface preparation protocols–including precision machining and controlled micro-roughening–will be developed to create a uniform interface for sensor attachment without introducing stress concentrators (Fig. 5-Step 2). These refinements will ensure that the AM-built substrate provides a stable foundation for subsequent embedding steps.

Sub-Task 2.B – Sensor Integration and High-Temperature Encapsulation. The team will develop a standardized workflow for integrating commercial RTDs, strain gauges and piezoelectric transducers sensors onto the AM surface (Fig. 5-Step 3). Flexible printed-circuit packaging, spot-welded electrical connections, and high-temperature structural adhesives will be optimized to ensure mechanical stability and electrical reliability. A high-temperature ceramic shielding layer will then be deposited to protect the sensors from thermal gradients during over-deposition (Fig. 5-Step 4). Deposition thickness, curing conditions, and adhesive formulations will be tuned to balance thermal isolation, stiffness, and compatibility with subsequent metallization.

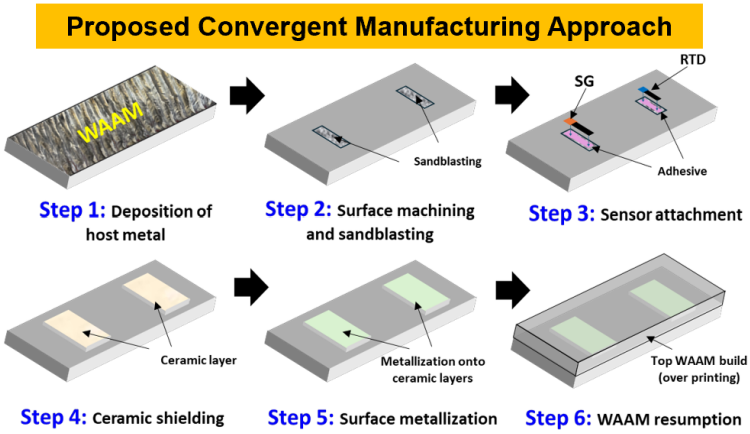

Fig 5. Proposed convergent manufacturing pathway illustration of the overall sensor embedding approach into metallic structures.

Sub-Task 2.C – Conductive Metallization for Electrical Continuity. This sub-task will focus on developing a conductive metallization layer that restores electrical conductivity over the ceramic shield and enables stable AM resumption (Fig. 5-Step 5). The team will identify optimal powder blends, deposition pressures, and tool-path strategies to achieve a dense, conductive coating with strong interfacial adhesion. The metallized layer will be engineered to be minimally thick while providing sufficient conductivity and mechanical anchoring for subsequent AM deposition.

Sub-Task 2.D – AM Resumption. The team will develop print-planning strategies for resuming AM deposition over the metallized sensor region (Fig. 5-Step 6). Pre-heating protocols, dynamic wire-feed adjustments, and real-time thermal monitoring will be used to ensure strong bonding between the metallized layer and the subsequently deposited AM layer. This sub-task will establish the conditions required for seamless integration of embedded sensors within the structure.

Task 3 – Simulation-Guided Sensor Network Design and Integration (Joint)

This task will develop a compact, manufacturing-aware simulation and optimization capability to design embedded sensor networks that satisfy sensing requirements while limiting structural penalty. The effort couples multi-physics modeling, progressive-damage analysis, and multidisciplinary optimization to produce manufacturable network configurations for prototype fabrication and qualification.

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Sub-Task 3.A – Multi-physics model development. Develop reduced-order, multi-physics finite-element model (FEM) that represent coupled thermo-mechanical behavior of AM, parts with embedded adhesives, ceramic shields, cold-spray layers, and wiring. Models must predict local temperature variation, interfacial stresses, and overall structural response for prospective network geometries while maintaining computational feasibility.

Sub-Task 3.B – Thermal survivability and sensing fidelity analysis. Perform transient thermal analyses to establish allowable thermal exposure, ceramic thickness ranges, and metallization conductivity requirements that preserve sensor and wiring integrity and limit temperature measurement error during deposition cycles and post-processing.

Sub-Task 3.C – Structural impact and progressive-damage assessment. Quantify effects of cavities, channels, and embedded layers on stiffness, peak stresses, and fatigue initiation using static, dynamic, and progressive-damage simulations. Derive geometric limits and design rules that constrain mechanical debit and inform acceptable tradeoffs between sensing coverage and structural performance.

Sub-Task 3.D – Topology and network design optimization. Execute a two-stage optimization: Stage-1 selects sensor count and coarse locations to maximize coverage and observability; Stage-2 refines cavity and channel geometries to minimize structural impact subject to manufacturing constraints from Task 2. Multi-fidelity and reduced-order strategies will be employed to control computational cost and accelerate design iteration.

Sub-Task 3.E – Model calibration and manufacturability validation. Model calibration will use experimental outputs from Task 2 to reduce uncertainty in thermo-mechanical response and interfacial behavior. Measured thermal histories, material property data, and inspection results will be incorporated into the simulation framework to update material models, cohesive laws, and boundary conditions through inverse analyses and reduced-order parameter estimation, ensuring that FEM predictions reproduce key observables within defined tolerances. The calibrated models will then be used to prioritize manufacturable sensor-network configurations for Task 4 prototype fabrication and validation. Figure 6 illustrates this workflow: the top panel shows representative network topologies (sensor types, cavity footprints, and wiring routes), while the bottom panel presents an indicative FEM tension comparison between a pristine 6 mm plate and an identical plate containing 2 mm-deep sensor cavities (4 × 4 mm) and 2 mm-wide wiring channels. This example indicates only a 2% increase in peak von Mises stress and a 0.2% decrease in elastic modulus, supporting feasibility. Quantitative metrics extracted from the calibrated models—including local thermal excursions, percent stress change, and predicted measurement error—will establish acceptance thresholds for optimization and determine which designs advance to Task 4 testing.

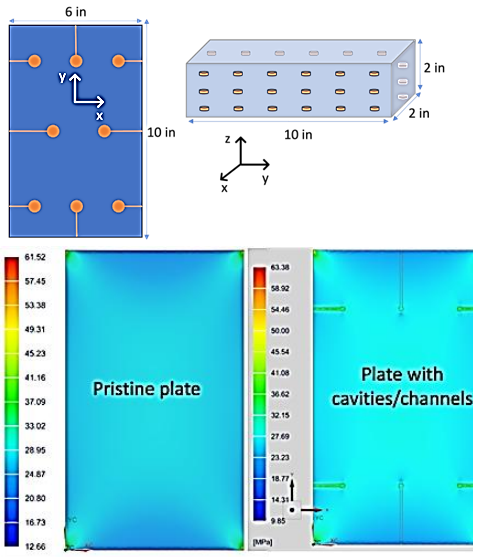

Fig.6 (Top) Illustration of sensor network topology, (Bottom) preliminary FEM results for pristine thin plate (i.e., baseline with no sensor networks) and a plate with 2mm-deep sensor cavities and channels. A 2% increase in the max Von-Mises stress under tensile load is observed.

Task 4 – Prototype Fabrication & Validation (Joint)

Task 4 will fabricate application-relevant prototypes using the convergent manufacturing pathway established in Task 2 and the optimized sensor-network designs produced in Task 3. The resulting smart metallic structures will undergo comprehensive mechanical and functional validation to assess sensing fidelity, structural performance, durability, and reliability. Final prototype geometry and use-case requirements will be co-defined with the Korean partners, ensuring relevance to their operational needs. Industry collaborators will support benchmarking against existing commercial solutions to evaluate competitiveness and transition potential.

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Sub-Task 4.A – Manufacturing Process & Sensor-Integration Assessment. This subtask will verify the reliability of the convergent manufacturing sequence and the survivability of embedded sensors after each processing step. Baseline AM specimens (no sensors embedded) will first be characterized using tensile tests to establish anisotropic mechanical properties. Static tensile tests for strain sensors and transient/steady-state thermal measurements for RTDs will confirm sensor functionality and identify integration issues. **Dolphicam ultrasonic NDE** will be used after each embedding step to generate high-resolution diagnostic images of sensor locations, wiring channels, and surrounding material. These scans will identify manufacturing-induced defects such as voids, porosity, incomplete bonding, and local delamination. Representative images in Figure 7 illustrate the defect morphology, depth, and spatial distribution, providing a ground-truth assessment of structural integrity. Quantitative metrics extracted from these scans will be used to (i) verify successful sensor encapsulation, (ii) diagnose integration failures, (iii) refine WAAM/ceramic/cold-spray parameters in Task 2, and (iv) update interfacial and defect-growth assumptions in the Task 3 simulation framework.

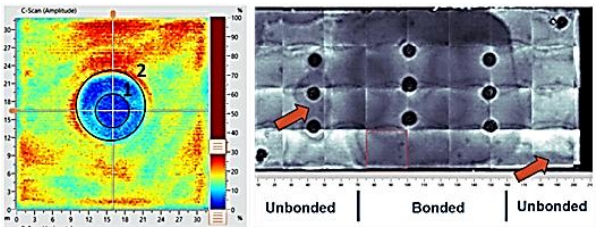

Fig.7 Representative Dolphicam NDE diagnostic results for an aluminum interface with bonded and unbonded areas.

Sub-Task 4.B – Static & Quasi-Static Validation. Static and quasi-static experiments will quantify sensing performance and structural response under controlled loading. Tensile tests and 3-point bending tests will be performed on baseline and embedded-sensor specimens to evaluate modulus, strength, ductility, and strain/temperature measurement accuracy. Testing will begin at room temperature and extend to elevated temperatures up to **350°C**. Thermal transients will be imposed using hot-plate heating and thermal-chamber, followed by steady-state holds to compare measured temperature fields with Task 3 predictions. Dolphicam scans before and after loading will identify damage initiation (debonds, cracks) and correlate NDE signatures with sensor outputs and FEM predictions.

Sub-Task 4.C – Dynamic (Fatigue) Validation. Fatigue experiments will evaluate long-term durability of embedded sensors, wiring, and interfaces. Tests will mimic **ASTM E466-21, ASTM E467-21, and ASTM E468/E468M-23**, using controlled stress amplitudes and frequencies until failure. Multiple baseline and embedded-sensor specimens will be tested to establish statistical confidence. Periodic Dolphicam scans will track defect evolution and identify failure modes. Objectives include quantifying fatigue debit, assessing sensor lifetime, and determining whether cavity geometry or routing rules require refinement. Testing will be conducted at room temperature and at elevated temperatures representative of the intended operating envelope.

Task 5 – TRL advancement and transition planning (joint)

Task 5 will integrate technical results from prototype fabrication, validation, and modeling to assess system readiness against TRL-5 expectations and identify the scale-up, qualification, and manufacturability gaps that must be addressed for future deployment. Working jointly with the Korean partners and industry collaborators, the team will define the next-phase commercialization pathway, including pilot-line demonstrations, material-system extensions, and benchmarking strategies. By structuring the effort around a partner-aligned transition plan, Task 5 positions the program as a **translation platform**-a capability that evolves toward industry relevance through coordinated technical maturation and co-engineered application definition.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The broad scope of the proposed convergent manufacturing effort requires a diverse team of experts spanning multiple domains. **Prof. Johnson Samuel** will serve as the project lead, coordinating overall execution with the Korean team and leading technical meetings and joint activities, with a focus on AM, machining, and convergent manufacturing. **Prof. Semih Akin** brings expertise in surface engineering (i.e., conductive surface metallization), ceramic processing, and direct metallization workflows. **Prof. Fotios Kopsaftopoulos** will contribute to structural health monitoring and experimental mechanics, leading static, dynamic, and environmental testing as well

	as computational modeling of embedded sensing systems. **Prof. Sandipan Mishra** specializes in process monitoring and control and will lead real time control strategies for WAAM, metallization, and ceramic deposition processes. **Prof. Daniel Lewis** will bring expertise in materials characterization and high temperature testing, overseeing microstructural, thermal, and mechanical evaluations through the Materials Research Center. The research team will be supported by two Ph.D. students and two undergraduate researchers who will carry out experimental, modeling, and integration tasks. RPI's **Center for Smart Convergent Manufacturing Systems (CSCMS)** provides all required manufacturing capabilities—including a robotic WAAM cell with coordinated multi robot operation, a FormAlloy system for rapid alloy development, an AFP platform, and a Mikrotools hybrid micro machining center—along with a robust convergent manufacturing control architecture for real time sensing and tool tip based correction. Additional resources include Prof. Kopsaftopoulos' **Intelligent Structural Systems Lab**, equipped with a 100 kN dynamic testing frame, environmental chamber, NI data acquisition systems, and a Dolphicam ultrasound imaging system, as well as the RPI's **Materials Research Center (MRC)** for high temperature mechanical and microstructural characterization. Together, these facilities provide comprehensive support for the proposed research and ensure seamless collaboration with **Korean partners**.
Expected R&R of Korean SME Partner	The Korean SME partner will play a critical role in translating RPI's patent-pending sensor-embedding technology into industry relevant additive manufacturing applications relevant to smart metallic structures. Their responsibilities will focus on defining end use requirements, validating performance under operational conditions, and supporting commercialization pathways. Specifically, the SME will: • **Define application requirements and constraints** by providing industrial use cases, product specifications, and operating envelopes that guide the adaptation of sensor embedded structures to real market needs. • **Provide validation standards and performance benchmark**s by aligning prototype testing with industry qualification protocols, certification requirements, and customer expectations. • **Participate in prototype testing and validation** through joint demonstrations, environmental exposure studies, and mechanical/thermal performance assessments conducted in collaboration with RPI facilities. • **Support commercialization and follow on development** by integrating prototypes into industrial workflows, advising on supply chain readiness, and contributing to techno economic analysis for scale up. • **Offer commercial feasibility input** by identifying market adoption challenges, cost constraints, and scalability expectations within Korean manufacturing ecosystems. • **Provide feedback on prototypes and deliverables** to ensure alignment with industry standards and readiness for pilot production. Through these roles, the Korean SME partner will ensure that the joint R&D effort advances beyond laboratory validation toward application-relevant prototypes, enabling TRL progression and accelerating the path to commercialization.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	This project is expected to establish a 'high-risk, high-reward' additive manufacturing collaboration that enables sensor-embedded smart metallic structure prototypes by advancing technology readiness levels and reducing risk for industrial deployment. The research is anticipated to deliver significant scientific, technological, and industrial advancements in smart structural products. The project will deliver outcomes with the following anticipated impacts: • **Technical Outcomes:** Application-specific prototypes with embedded sensing, demonstrated TRL advancement to 4–5, and validated performance datasets ensuring fidelity under mechanical, thermal, and environmental conditions. • **Commercialization Potential:** Reduced technical risk for smart structural products, differentiated value-added components, and a clear pathway to scale-up and deployment within Korean SME manufacturing ecosystems. • **In-field testing and deployment:** Execution of pilot-scale field tests on relevant operational environments to verify durability, sensing accuracy, and system integration, providing critical data for pre-commercial validation. The application fields include, but are not limited to, **aerospace**, **marine**, **civil infrastructure**, and **nuclear power**. • **Broader Impact:** Strengthened U.S.–Korea research collaboration, enhanced competitiveness

	of Korean SMEs through access to patented sensor-embedding technology, and workforce development via graduate and undergraduate training with knowledge transfer across institutions.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization